DATA AGENTS

Data Agents in Microsoft Fabric

Samantha Cruz

DATA SATURDAYS SOFIA

10 OCTOBER 2026 | SOFIA

DATA SATURDAYS

SOFIA
Microsoft
Fabric User Group

Microsoft

OUR AMAZING SPONSOR

PREDICTA
ANALYTICS FOR DECISION MAKING

<epam>

ALWAYS SOFT PROJECT BULGARIA

MVP Microsoft® Most Valuable Professional

KINKAO _SOFIA

SOFIA TECHPARK

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MEET THE SPEAKER



SAMANTHA CRUZ

Data Engineering Team Lead at Isolutions

DATA SATURDAYS

10 OCTOBER 2026

At Sofia Tech Park

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WHAT WE'LL COVER TODAY

01

What & how

The building blocks and what happens between question and answer

02

Build & activate

Creating one, and the tenant setting everyone forgets

03

Fresh from FabCon

Topics, ontologies, Fabric IQ, MCP: GA vs preview vs not yet

04

Trust & limits

Evaluating, monitoring, and what it still can't do

05

The bigger picture

Copilot, the new agent family, and the rest of the industry

06

Live demo

Building an agent on real e-commerce data

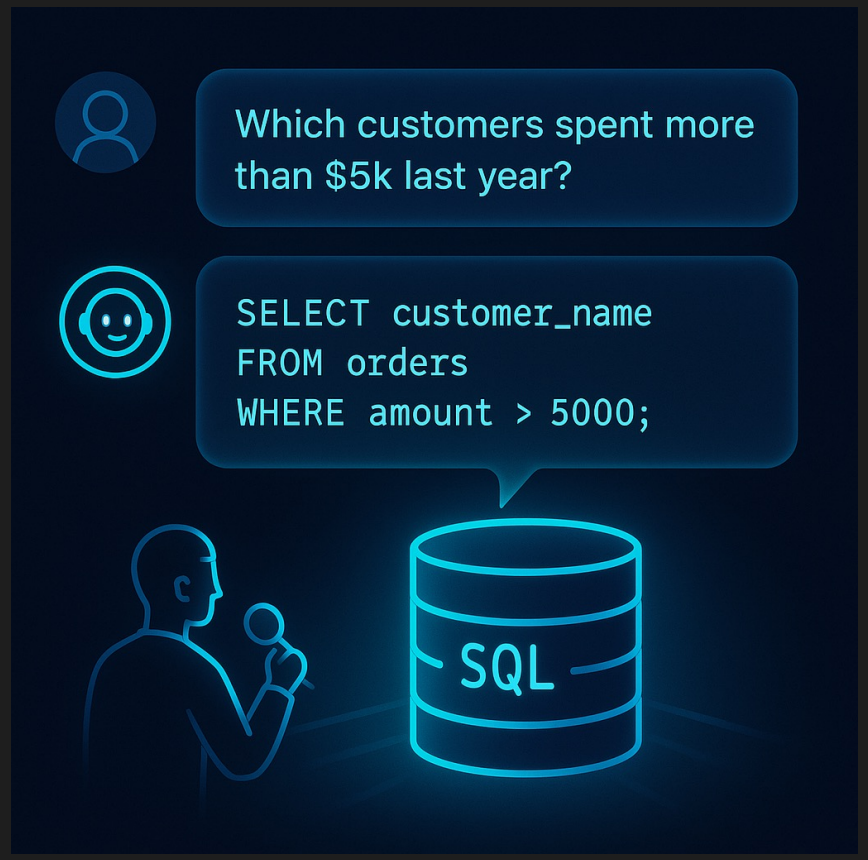
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THE PROBLEM WE'RE ACTUALLY SOLVING

TODAY

1. Business user has a question
2. Pings an analyst on Teams
3. Analyst context-switches, writes SQL/DAX
4. Answer arrives… other tasks delayed
5. Question #2 restarts the whole loop

WITH A DATA AGENT



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WITH A DATA AGENT

1. Business user asks in plain English
2. Agent checks permissions, same rules as always
3. Translates the question to SQL / DAX / KQL
4. Answer comes back in seconds
5. No analyst needed for the routine stuff

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SO, WHAT IS A DATA AGENT?

GA · MAR 2026

A Fabric item that lets people have natural-language conversations with enterprise data, without ever showing them SQL, DAX or KQL.

Built on Azure OpenAI.

CONVERSATIONAL

Plain-English questions in, structured answers out

AN ABSTRACTION LAYER

Sits over your query engines, doesn't replace them

PERMISSION-AWARE

Inherits exactly the access you already have, no shortcuts

GENERALLY AVAILABLE

No longer a preview toy: GA since March 2026

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FROM QUESTION TO ANSWER

Ask
Plain-English question

Validate
Permissions & security context

Identify
Best-fit data source

Translate
To SQL / DAX / KQL

Answer
Read-only, human-readable

“If the underlying data model is messy, the agent's answers will be messy too.”

Garbage in, confidently-worded garbage out.

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WHICH MODEL REASONS OVER YOUR QUESTION?

Microsoft doesn't publish the exact model name.

Data agents use large language models through Azure OpenAI.

The same LLM:

→ interprets the natural-language question

→ reasons over the schema

→ decides which query type to generate

→ writes the final answer

BUT YOU CAN CHOOSE THE RUNTIME

GA · DEFAULT

Standard runtime

Stable and predictable, for production. Supports up to 25 tables per data source.

PREVIEW

Preview runtime

Newer model for SQL generation, smarter DAX, asks when a question is ambiguous. Needed for Topics, schema descriptions and more than 25 tables.

Same base LLM on both runtimes. A published agent keeps its runtime until you republish.

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WHICH MODEL REASONS OVER YOUR QUESTION?

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name

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→ deci

→ write

Data Agent orchestrator upgraded to GPT 5.1

The data agent orchestrator now runs on GPT-5.1, across both the standard and preview runtimes. The orchestrator handles how questions are rephrased, how work is planned across your data sources, and how the final answer is composed — so this upgrade changes behavior in all three. Most of what you see should be an improvement in answer quality and planning, but the change is not behavior-neutral: prompts tuned against the previous model may produce different results. We recommend re-running your evaluations, reviewing the results against your saved baselines, and updating your agent instructions and prompts where the new behavior doesn't match what your scripts expect.

To learn more, refer to the data agent runtimes documentation.

tables

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and more than 25 tables.

August 2026

Same base LLM on both runtimes. A published agent keeps its runtime until you republish.

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WHICH MODEL REASONS OVER YOUR QUESTION?

Microsoft doesn't publish the exact model nam

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→ writes the final answer

BUT YOU CAN CHOOSE THE RUNTIME

Model Upgrades on the Data Agent Runtime (Preview)

The Fabric data agent runtime (Preview), now runs on the newer 5.6 model, delivering improvements in response quality, accuracy, and latency across the end-to-end experience. The upgrade powers the agent's orchestrator—including question rephrasing and planning—as well as source routing, Code Interpreter, natural-language querying for Power BI semantic models, and natural language to SQL.

es

Preview runtime

Newer model for SQL generation, smarter DAX, asks when a question is ambiguous. Needed for Topics, schema descriptions and more than 25 tables.

September 2026

Same base LLM on both runtimes. A published agent keeps its runtime until you republish.

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PREVIEW RUNTIME: WHAT EXACTLY IMPROVES

Source	Improvement	What it does
SQL	Schema descriptions	Business context on tables and columns for large or ambiguous schemas
SQL	Topics	Up to 1M characters of guidance; the agent picks the relevant sections
SQL	Newer SQL model	Advanced NL2SQL: more accurate queries, lower latency
SQL	Follows your examples	Sticks to your example query patterns, no invented filters
SQL	Implied filter values	Works out filter values the user only implied
SQL	Ambiguity handling	Asks a clarifying question instead of guessing
DAX	Multi-step reasoning	Advanced DAX: checks results and refines for complex questions
DAX	Reliable filters	Looks up real column values before filtering

Switch it on: agent settings, or the SDK: `update_configuration(enable_preview_features=True)`. Then republish.

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ONE AGENT, MANY DATA SOURCES

Source	Artifacts	Query language	Status
SQL	Lakehouse, Warehouse, SQL Database, Mirrored DBs	T-SQL	**GA**
Eventhouse	KQL Database	KQL	**GA**
Semantic model	Power BI semantic models	DAX	**GA**
Graph	Fabric Graph model	GQL	**Preview**
Ontology	Fabric IQ ontology, now also agent context	Ontology-native	**Preview**
Azure AI Search	AI Search index (PDFs, text, docs)	NL + search	**Preview**

Read-only, live, no copies. It reads what's already there and you can combine sources in one agent. Schema-size limits were lifted: 1,000+ tables are fine.

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MIN. F2 CAPACITY

CREATING A DATA AGENT

1

Define context

What is this agent for?
Domain, tone, scope

2

Select source(s)

Lakehouse / Warehouse / semantic model…

3

Add examples

Question → query pairs for the tricky logic

4

Test it

Ask real questions before you publish

PREVIEW

Build agent with AI mode

Point it at semantic models, documentation, glossaries and data dictionaries. It drafts instructions, examples and config for you. Less manual setup, richer agents.

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TENANT SETTINGS

Data agents rely on tenant-wide settings: Copilot / Azure OpenAI must be enabled for your tenant (or the right security group).

Data residency

Cross-geo processing settings can block it entirely

Compliance

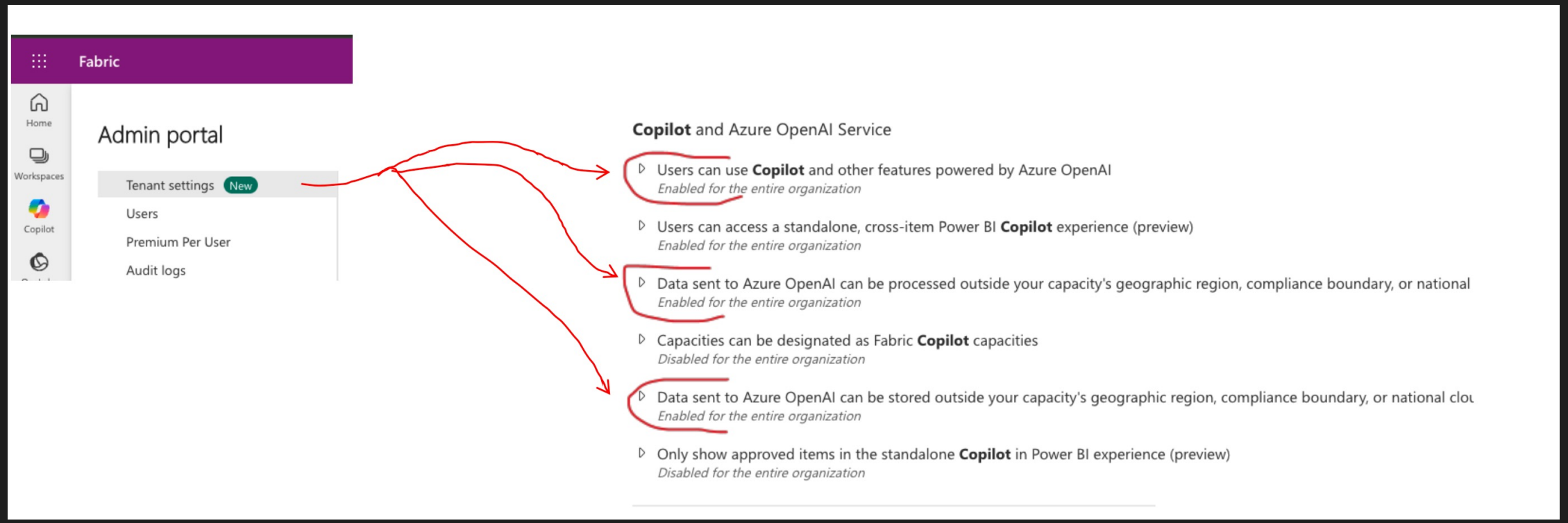
Security and DLP first. Sensitivity labels carry over to Copilot

IT conversation

A tenant-admin decision, not a workspace one

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TENANT SETTINGS



This setting is only applicable for customers who want to use Fabric data agents whose capacity's geographic region is outside of the EU data boundary and the US.

https://learn.microsoft.com/en-us/fabric/data-science/data-agent-tenant-settings



DATA SATURDAYS

FRESH FROM FABCON EUROPE

Barcelona · 28 Sep – 1 Oct 2026

GA + DATE

PUBLIC PREVIEW

NOT PUBLIC YET

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TOPICS: STRUCTURED GUIDANCE PER SOURCE

PUBLIC PREVIEW · SEP 2026

1M

characters of guidance per supported data source

Way beyond what fits in agent instructions: business definitions, tables, relationships, filters and calculation rules.

Requires the Preview runtime

TOPIC Sales

“Revenue = net of returns, excl. VAT. Fiscal year starts in July.”

TOPIC Inventory

“Stock-out = on_hand < safety_stock for 2+ days.”

TOPIC Customer retention

“Churned = no order in 180 days. Exclude B2B test accounts.”

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ONTOLOGIES: YOUR BUSINESS, EXPLAINED

PREVIEW · SEP 2026

Ontology as data-agent context

Grounds query generation in enterprise entities, relationships, definitions and source bindings, for more accurate, more explainable answers.

PREVIEW · SEP 2026

Ontology as Real-Time Dashboard source

Business entities and relationships straight into live visuals.

ANNOUNCED · NOT PUBLIC YET

Agentic ontology creation + DAX measures

Build ontologies from semantic models, OneLake data and documents, and bring Power BI DAX measures in. Announced at FabCon EU.

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ONE AGENT, MANY WAYS TO USE IT

GA · 28 SEP 2026

Microsoft Copilot Studio

The GA route into Teams and Microsoft 365 Copilot

GA · 28 SEP 2026

Visualizations in answers

Charts in the reply. Goodbye “can't do visuals”

GA

Copilot in Power BI

Discover and chat with data agents in Power BI

GA

Azure AI Foundry

Plug the agent into your own multi-agent apps

GA

Python SDK + MCP

Programmatic access; MCP-capable clients

GA · SEP 2026

Microsoft 365 Copilot

Via Fabric IQ: business users never leave Copilot

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WHOSE CREDENTIALS DOES THE AGENT USE?

BY DEFAULT

The user's own identity

Every query runs as the person asking, in Fabric, Teams, Copilot Studio or M365 Copilot.

✓ RLS, CLS and Purview policies apply

✓ Read on a semantic model is enough: no Build, no workspace access

! RLS doesn't apply to workspace Admins, Members or Contributors. Test with a viewer.

IF USERS SHOULDN'T TOUCH THE DATA

COPILOT STUDIO

Agent author authentication

Runs with the author's credentials. Users need no data access, but everyone sees what the author sees. Teams only.

NO RLS UNTIL ~Q1 2027

Service principal (your own app)

The app's identity reads the data. No per-user RLS yet, and no KQL sources.

RECOMMENDED

Curated model + Read only

Aggregated tables with RLS/OLS, users get Read without Build: no Excel, XMLA or new reports.

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TRUST, BUT VERIFY: EVALUATING ANSWERS

```
from fabric.dataagent.evaluation import *
eval_id = evaluate_data_agent(df, agent)
summary = get_evaluation_summary(eval_id)
```

index	evaluation_id	true_count	false_count	unclear_count	true_percentage
0	2cf897c6-801c-4e94-8664-f4848ffb6071	3	1	0	75.0
1	30c590da-6df8-494f-a359-642d34d4fa32	2	2	0	50.0
2	34fc1c4c-8696-48e5-aebb-78b3b681cc93	2	2	0	50.0
3	3879189e-f216-4f43-ae2e-9f96933a1429	2	2	0	50.0
4	45dfc4ae-a6ae-4265-80ff-a75c7b463fb0	4	0	0	100.0
5	726cb9fe-614a-4dc0-b121-18dd224aa391	2	2	0	50.0
6	75d8e7db-3448-4e93-bb20-819e038e28a6	4	0	0	100.0
7	760af511-c8a8-497c-b729-6b113f4ac2a9	3	1	0	75.0
8	76e1e1c8-29ee-4505-bce2-9cd331927ef4	3	1	0	75.0
9	7b65fc7b-c8c8-46a1-9c2b-04adf65f4f48	3	1	0	75.0

Why bother?

Catch bad answers before your users do, and re-test after every config or model change.

Keep in mind

Batch, not real-time. Python SDK only. Evaluating never changes the agent.

PREVIEW · SEP 2026

Feedback in Microsoft 365 Copilot

Creators review consumers' thumbs-up/down to improve answers.

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MONITORING: WHO ASKED WHAT?

TRY IT TODAY

PUBLIC PREVIEW

Monitor hub: Agents page

Activity and health metrics for every data agent and operations agent you can see. No opt-in needed.

PUBLIC PREVIEW

Purview audit (DSPM)

Every prompt and response, including generated queries, in the Microsoft 365 unified audit log.

ON THE ROADMAP

PLANNED · PREVIEW Q4 2026

Creator feedback loop

Creators see how consumers actually use their agents: questions, answers, satisfaction.

PLANNED · PREVIEW Q4 2026

Agent 365 identity & observability

Data agents get an enterprise identity in Agent 365 and send activity telemetry to IT and security.

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WHAT A DATA AGENT CAN'T DO (YET)

STILL CAN'T

– No writes: can't create, update or delete

– No hand-tuned SQL joins

– Limited control over the generated query

NOW IT CAN

✓ Visualizations in answers (GA · 28 Sep 2026)

✓ No more 25-row cap: up to 1,000 rows (preview · Sep 2026)

✓ Production use: GA since Mar 2026

STILL YOUR JOB

– Data governance

– Data quality: it won't fix bad data

– Tenant settings & capacity

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A FEW THINGS THAT MAKE A HUGE DIFFERENCE

01

Name things clearly

Clear table/column names help the agent

02

One domain per agent

Specialize, no do-everything agents

03

Minimize scope

Only the tables you need answered

04

Say what TO do

Positive instructions beat “don't” lists

05

Define your jargon

Business terms, abbreviations, synonyms

06

Use example queries

Show the joins/filters too complex to guess

07

Use Topics (new)

Move big guidance out of the instructions

08

Evaluate every change

Re-run the SDK after each tweak

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DATA AGENT VS. COPILOT VS. COPILOT STUDIO

	Copilot in Fabric	Fabric Data Agent	Copilot Studio
Scope	Single-step, in-tool help	Multi-step, standalone, governed Q&A	Orchestrates many agents/tools
Memory	Session-bound	Keeps workspace context across runs	Depends on the orchestrator
Best for	Iterating inside a report	Recurring, governed Q&A	Cross-system scenarios (Teams, M365)

Copilot answers: **“help me write this query.”**

Data Agent answers: **“what was our churn last month?” using the lakehouse.**

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THE FABRIC AGENT FAMILY IS GROWING

PREVIEW · FABCON EU

Data engineering agent

Built on the Osmos acquisition. Long-running work: migrations, lakehouse modernization, performance tuning, complex ETL. Works from GitHub Copilot, VS Code and Claude Code.

PREVIEW · FABCON EU

Operations agent

Watches real-time data and pipelines. New: pre-approved actions and root-cause analysis for failures.

NOT PUBLIC YET · SOON

Database agents

For SQL and PostgreSQL on Azure: investigate issues, find bottlenecks and optimization opportunities, via the new Database Hub.

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EVERY PLATFORM HAS BUILT THIS NOW

Platform	Their equivalent	How it works
Microsoft Fabric	Data Agent + Fabric IQ	NL → SQL/DAX/KQL, read-only, Purview-governed
Databricks	Genie agent	NL → SQL over Unity Catalog; managed MCP server
Snowflake	Cortex Agents / Snowflake Intelligence	Fuses structured (Analyst) + unstructured (Search)
Google Cloud	Conversational Analytics	Gemini over BigQuery + Looker
AWS	Amazon Q in QuickSight	NL prompt → full dashboard, deep AWS-native
ThoughtSpot	Spotter agents	NL search-to-insight

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SAME PATTERN, DIFFERENT BACKBONE

THE MECHANISM CONVERGES

LLM + governed semantic layer → NL to query → read-only, permission-scoped.

THE MOAT IS GOVERNANCE

Fabric: Purview + OneLake + Fabric IQ. Databricks: Unity Catalog.

PICK BY METADATA, NOT HYPE

Choose the agent that lives where your semantic layer already lives.

Not “best agent wins”, but “who already owns my metadata”.

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2026 AT A GLANCE

MAR	JUN	JUL	SEP	28 SEP	NEXT
Data agents GA	Service principals	Local MCP	Topics	Copilot Studio	Database agents
Fabric Remote MCP	Data Agent Runtime		25 → 1,000 rows	Visualizations	IQ in Copilot Code
			Ontology context	Fabric IQ in Copilot	DAX in ontology
			Advanced NL2SQL	IQ + Core MCP	A365 observability
			M365 feedback	AI mode	Creator feedback loop

GA + DATE | PUBLIC PREVIEW | NOT PUBLIC YET

Mar 2026 = month only (Fabric roadmap, Q1). 28 Sep 2026 = FabCon Europe announcement day.

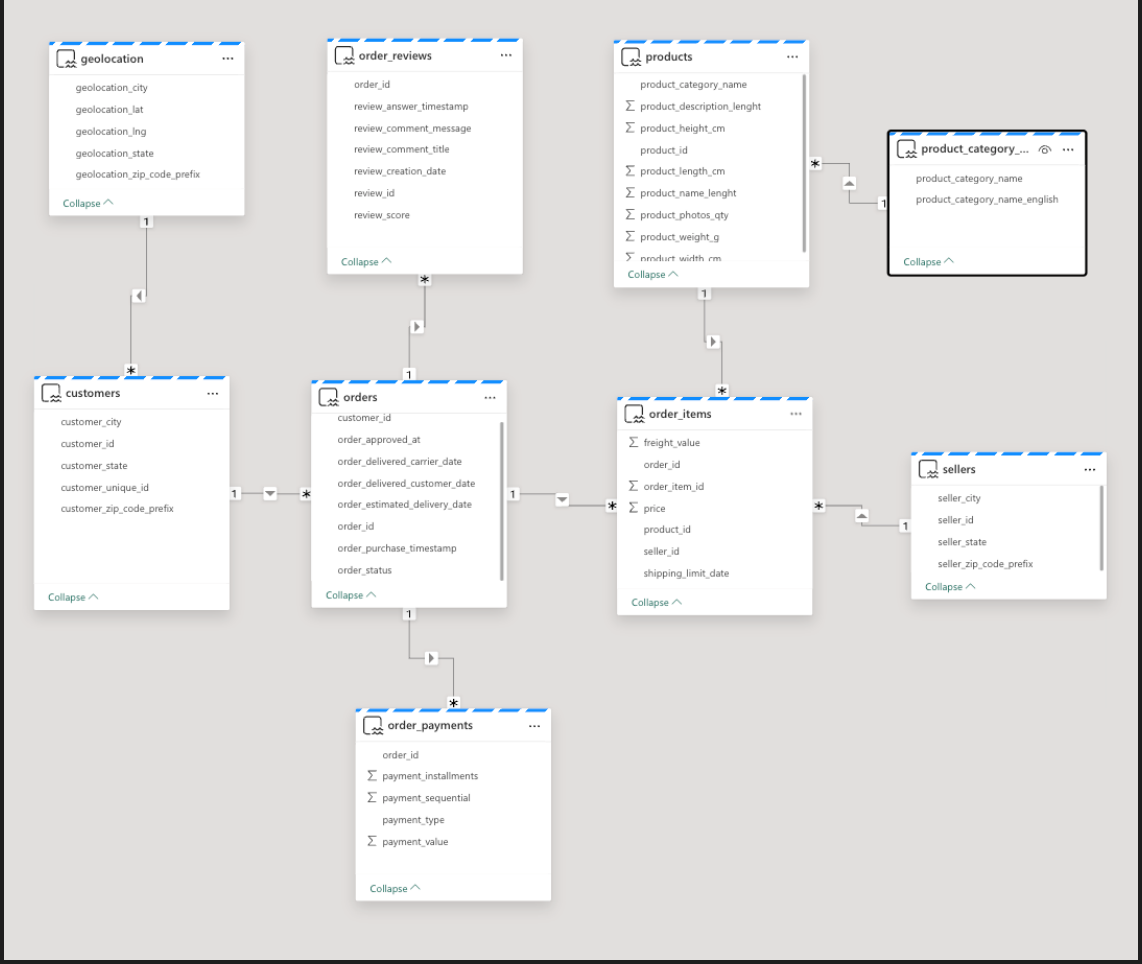
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D E M O

OVERVIEW



THE DEMO DATASET: OLIST E-COMMERCE



Real e-commerce data

9 related tables: customers, orders, payments, reviews, products, sellers…

Gold layer, already modeled in the Lakehouse

Why it works

Genuine relationships + clear business questions: exactly what a data agent needs to shine.

Source: kaggle.com/datasets/olistbr/brazilian-ecommerce



DATA SATURDAYS

THANK YOU

Feel free to ask any questions

No need to be shy

Maybe you'll get the perfect thing

See you next year 2027

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